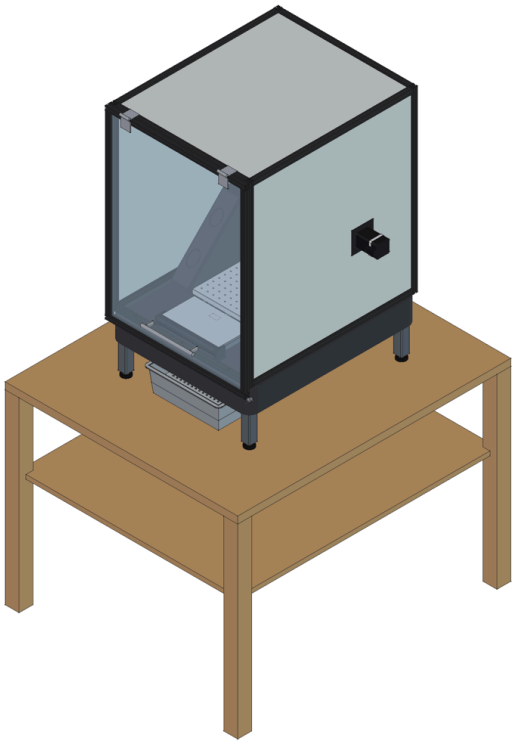
Build guide 5: enclosure, coolant and commissioning

DIY CNC epoxy-granite mill · October 2026

8. Enclosure and coolant (part 3 of the sourcing list)

The machine runs without these: until then, brush or drip cutting fluid onto the cut, keep the way covers and screw cover on, and let the gutters and drain carry the fluid to the pan under the base. Cast the enclosure inserts into the base now so the frame bolts straight on later.

1. Bolt the 2020 frame to the base top with angle brackets into the cast-in M5 inserts (M5 x 35). Cut list in `docs/5-design/report.txt`.
2. ACM side, back and roof panels; polycarbonate door on hinges and gas struts, magnetic catch. Brush seals where the X and Y motors pass through the panels.
3. Way covers: the stainless strips on their spring rollers in front of and behind the table, the spiral cover on the Y screw, the drip shield over the X rails.
4. Coolant: slide the sump (steam-table pan with the chip basket, magnet, baffle, screen and 24 V pump) in under the base below the drain nipple. Run the return hose to the bulkhead, up the cable chain, to the manifold and nozzles on the spindle clamp. Use water-soluble coolant.



9. Commissioning

1. **Run each axis by hand,** then with the motors at low speed, end to end. Listen for tight spots.
2. **Homing and soft limits** in FluidNC; check the travel (180 / 300 / 140 mm).
3. **Steps per mm:** move 100 mm and measure with the dial indicator or a caliper; correct.
4. **Squareness:** sweep an indicator in the spindle along a straightedge or the table edge. X to Y is set by the beam position (step 4 of the frame); Z to the table is set by shimming the spindle clamp (tram). Aim for under 0.02 mm over 100 mm.
5. **Face the table:** take a light fly-cut over the aluminum table so it is flat and parallel to X/Y.
6. **First cuts:** aluminum first, then steel at light depths of cut with flood coolant. Check the spindle temperature and the coolant flow, and re-torque the frame bolts and the riser hold-downs after the first few hours of running.